Important

New for TJ, the Powertrain Control Module and Transmission Control Module have been combined into single control module on all vehicles equipped with a 2.4L and 4.0L engines. This new module is the Next Generation Controller (NGC). New Diagnostics procedures along with new DTC's are two of the changes you will see which reflect the new combined module technology. The NGC modules will have four color coded connectors C1/A through C4/D, (C1-BLK, C2-ORANGE, C3-WHITE, C4-GREEN). Each PCM connector has 38 pins. Two new tools were introduced to help diagnose and repair the new PCM (NGC) terminals and harness connectors. The Miller #3638 terminal removal pick must be used to release the connector terminals, or harness and connector damage will occur. Also, the Miller #8815 Pin Out Box was introduced. You must use the Miller #8815 tool instead of probing the PCM terminals, or harness and connector damage will occur. There is also a new Verification test and module replacement procedure for the new PCM.

1.0 INTRODUCTION

The procedures contained in this manual include specifications, instructions, and graphics needed to diagnose the PCM Powertrain System. **The diagnostics in this manual are based on the failure condition or symptom being present at the time of diagnosis.** Please follow the recommendations below when choosing your diagnostic path.

1. First make sure the DRBIII® is communicating with the appropriate modules; i.e., if the DRBIII® displays a ″No Response″ condition, you must diagnose this first before proceeding.
2. Read DTCs (diagnostic trouble codes) with the DRBIII®.
3. If no DTCs are present, identify the customer complaint.
4. Once the DTC or customer complaint is identified, locate the matching test in the Table of Contents and begin to diagnose the symptom.

All component location views are in Section 8.0. All connector pinouts are in Section 9.0. All system schematic diagrams are in Section 10.0. All charts and graphs are in section 11.0.

An * placed before the symptom description indicates a customer complaint.

When repairs are required, refer to the appropriate service manual for the proper removal and repair procedure.

Diagnostic procedures change every year. New diagnostic systems may be added; current systems may be enhanced. READ THIS MANUAL BEFORE TRYING TO DIAGNOSE A VEHICLE DTC. It is recommended that you review the entire manual to become familiar with all new and enhanced diagnostic procedures.

After using this book, if you have any comments or recommendations, please fill out the form at the back of the book and mail it back to us.

1.1 SYSTEM COVERAGE

This diagnostic procedure manual covers Powertrain, Transmission, and Transfer Case diagnostics for 2005 TJ vehicles equipped with the Next Generation Control Module (NGC).

1.2 SIX-STEP TROUBLE SHOOTING PROCEDURE

Diagnosis of the Powertrain Control Module (PCM) is done in six basic steps:

- verification of complaint
- verification of any related symptoms
- symptom analysis
- problem isolation
- repair of isolated problem
- verification of proper operation

2.0 IDENTIFICATION OF SYSTEM

The Powertrain Control Module (PCM) monitors and controls:

- fuel system
- ignition system
- charging system
- speed control system
- The NGC control module is used on the 2.4L and 4.0L engines with an automatic or manual transmission.
- The 42RLE transmission can be identified by confirming a Solenoid/Pressure Switch Assembly located on the right side of the transmission, The Transmission Range Sensor, Input Speed Sensor and Output Speed Sensor are located on the left side of the transmission. Refer to the Service Information for transmission ID tag descriptions.